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18/183,431	03/14/2023	Stefan Nordbruch	49010-924	1157

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EXAMINER
RHEE, ROY B

ART UNIT	PAPER NUMBER
3664	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/183,431	Applicant(s) Nordbruch et al.	
	Examiner ROY RHEE	Art Unit 3664	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 21 July 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2,4-6,10,12 and 15-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4-6,10,12 and 15-22 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 3/04/2023 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on April 24, 2026 has been entered.

Response to Amendment

Applicant's amendment filed on April 24, 2026 amends claims 1, 4, 10, and 12 and cancels claims 3 and 11. Claims 1-2, 4-6, 10, 12, and 15-22 are pending.

Response to Arguments

Applicant's arguments filed on April 24, 2026 regarding the newly presented claim limitations have been fully considered and are unpersuasive and/or moot as shown in the rejections under 35 U.S.C. 103 that follow. The newly presented

limitations in the independent claims are taught by the previously cited references, Kundu and Das.

Drawings

The drawings are objected to under 37 CFR 1.83(a) because the flowchart boxes in at least Figs. 1, 6-7 and 9-11 lack descriptive text and therefore do not illustrate the steps recited in the claims. The drawings must show every feature of the invention specified in the claims. Descriptive text must be shown in each of the steps of the flowchart boxes. No new matter should be entered.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as “amended.” If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either “Replacement Sheet” or “New

Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Objections

Each of claims 1 and 10 is objected to because of the following informalities:

In claim 1, the words "... motor vehicle, it is checked ..." should be rewritten as "... motor vehicle, wherein it is checked ...".

In claim 10, the words "... over the communication link, it is checked ..." should be rewritten as "... over the communication link, wherein it is checked ...".

For each of claims 1 and 10, the words "... during log in whether ..." should be rewritten as "... during login whether ..." to be consistent with the specification.

For each of claims 1 and 10, the words "... presence of a predetermined safety integrity level (SIL or automotive safety integrity level ASIL) ..." should be rewritten as "... presence of a predetermined safety integrity level (SIL) or automotive safety integrity level (ASIL) ...".

The foregoing changes are required to correct clerical and/or grammatical errors.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claims 1-2, 4-6, 10, 12 and 16-21 are rejected under 35 U.S.C. 103 as being unpatentable over Kundu et al. (US 2022/0065644) in view of Das (US 2019/0283768).

Regarding claim 1, Kundu teaches a **computer-implemented method for at least partially automated driving of a motor vehicle, comprising the following steps: determining that a need exists for infrastructure-based, at least partially automated driving of the motor vehicle**; (see at least Kundu at [0030] in conjunction with Fig. 1, for example; see Kundu at [0033] which discloses that processor(s) 116 may be configured to fetch and execute computer-readable instructions stored in the computer-readable media 118, which may ***program the processor(s) to perform the functions*** described herein; see Kundu at [0028] which discloses that implementations herein may significantly improve the performance of a fully/***partially autonomous vehicle*** using connected data and that the connected vehicle is able to share data with the ***data analytics platform, which provides access to cloud databases and cloud computing power***, ***Web-based sources*** of information, as well as providing access to information provided by other vehicles; see Kundu at [0033] which discloses that ECU or other vehicle computing device 104 may include one or more processors 116, and that ***processor(s) 116 may be configured to fetch and execute computer-readable instructions stored in the computer-readable media 118, which may program the processor(s) 116 to perform the functions described herein***; see Kundu at [0041] which discloses that the vehicle control program 124 may send control signals to the suspension controller, the steering controller, and/or the vehicle speed controller for controlling or partially controlling the vehicle in some applications; see Kundu at [0055]

which discloses that the vehicle computing device 104 may ***request a route*** from the service computing device 108; see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for ***partially/fully autonomous vehicles***, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, ***infrastructure cameras and other sensors***, cellphones, other transportation data services, and so forth, as discussed above. Examiner notes that a need exists for infrastructure-based partially automated driving based on the vehicle computing device's request for a route from the service computing device. In other words, a request is made from the vehicle computing device to a service computing device because a need exists to make such a request.)

sending, in response to determining that the need exists for infrastructure-based, at least partially automated driving of the motor vehicle, a plurality of log-on data over a communication network to log on the motor vehicle to a remote infrastructure server and to establish a communication link between the infrastructure server and the logged on motor vehicle, it is checked during log in whether at least one safety condition for the infrastructure-based, at least partially automated driving of the motor vehicle is fulfilled, the at least one safety condition [is a presence of a predetermined safety integrity level (SIL or automotive safety integrity level ASIL) in the motor vehicle and in an infrastructure transmitting the infrastructure data**]** (see at least Kundu at [0038-0039] in conjunction with Fig. 1 which discloses that

the ***one or more networks 106*** may include any appropriate network, including a wireless network, such as a cellular network; a wide area network, such as the Internet; a local area network, such an intranet; a local wireless network, such as Wi-Fi; close-rang wireless communications, such as BLUETOOTH® or DSRC; a wired network, including fiber optics and Ethernet; any combination of the foregoing, or ***any other suitable communication network***. See Kundu at Fig. 1 elements 102, 104; Examiner maps element 102 to the motor vehicle. See Kundu at Fig. 1 element 108; Examiner maps service computing device(s) 108 to remote infrastructure server. See Kundu at [0095] which discloses that data analytics platform 145, the descriptive analytics module(s) 148 may decrypt the received vehicle data such as by using cryptographic hash algorithms such as MD5, SHA-1, SHA256, or other decryption techniques and that following decryption, the descriptive analytics module(s) 148 ***may authenticate*** or otherwise determine ***the identity of the vehicle and an occupant***, ***and may access the vehicle data database 162 to retrieve any information*** about the vehicle or occupant maintained therein. Kundu at [0095] further discloses that the examples of the information that may be retrieved may include the vehicle sensor configuration information 128 and/or ***vehicle configuration information 130***. Examiner notes that Kundu teaches retrieving ***any information (i.e., while authenticating or logging in)***, such as the identity of the vehicle and the occupant, by way of accessing the vehicle data database ***during authentication or log in***. Examiner maps authentication to log in.

Examiner further notes that determining the identity of the vehicle and an occupant corresponds to checking whether at least one safety condition for the infrastructure-based, at least partially automated driving of the motor vehicle is fulfilled.)

transmitting, via the communication network, a request to transmit a plurality of infrastructure data based on which the motor vehicle is drivable in an at least partially automated manner, in response to determining that a need exists for infrastructure-based, at least partially automated driving of the motor vehicle; (see Kundu at Figure 1, which illustratively depicts ***network 106 for communicating a request*** between vehicle computing device(s) 104 and service computing device(s) 108; see Kundu at [0055] which discloses that the vehicle computing device 104 may ***request a route*** from the service computing device 108; Examiner notes that a request is made based on the need for automated driving or route guidance by the vehicle computing device(s). Also, see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for ***partially/fully autonomous vehicles***, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, ***infrastructure cameras and other sensors***, cellphones, other transportation data services, and so forth, as discussed above. Examiner maps the route requested by the vehicle computing device(s), which utilize various different types of data from different sources such as vehicles, infrastructure cameras, and other sensors, to the recited infrastructure data.)

receiving, via the communication network, the infrastructure data in response to transmitting the request, based on which the motor vehicle is operable in the at least partially automated manner; (see Kundu at [0056] which discloses that upon determining the optimal route(s), the service computing device 108 may send the selected optimal route(s) 186 to the vehicle computing device 104; see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for ***partially/fully autonomous vehicles***, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, ***infrastructure cameras and other sensors***, cellphones, other transportation data services, and so forth, as discussed above; see Kundu at [0058] which discloses that the data analytics platform 145 may provide road anomaly and driver behavior information for analyzing and making determinations with respect to vehicle occupant safety and that the data analytics platform may share aggregated vehicle data. Examiner, for example, maps the sending of the selected optimal route(s) which incorporate the various sensor data to the receiving of the infrastructure data in response to the transmitting request.)

generating a plurality of control signals for at least partially automated controlling of a lateral and/or a longitudinal operation of the motor vehicle based on the infrastructure data; (see Kundu at [0041] which discloses that the vehicle control program 124 may determine an appropriate action, such as braking, steering,

accelerating, or the like, and may ***send one or more control signals to one or more vehicle systems 114*** based on the determined action. Also, see Kundu at [0041], which discloses that for example, the vehicle control program 124 may send control signals to the suspension controller, the steering controller, and/or the vehicle speed controller for controlling or partially controlling the vehicle in some applications. Examiner notes that control signals sent to the steering controller and/or the vehicle speed controller corresponds to controlling of a lateral and/or longitudinal operation of the motor vehicle.)

the control signals are generated based on a result of checking whether the at least one safety condition for the infrastructure-based, at least partially automated driving of the motor vehicle is fulfilled, (see Kundu at [0048] which discloses that examples of predictive analytics modules 150 may include destination prediction, candidate route prediction and monitoring, speed profile determination, and anomaly prediction, that examples of ***prescriptive analytics modules 152 may include modules for managing safety,*** efficiency, comfort, and the like of vehicles and/or vehicle occupants; see Kundu at [0071] which discloses that in some examples herein, rather than relying on driver response when lane departure occurs, the LKA system may employ sensor fusion from the long-range camera and the long-range radar to alert the driver and also ***activate the steering actuator***, that accordingly, the steering actuator may be ***automatically engaged to return the vehicle to its proper lane***, and that ***sensor***

fusion algorithms may be required to meet strict performance and ***safety requirements***. Examiner notes that sensor fusion data associated with the long-range camera and the long-range radar is a provided to the data analytics platform which is checked by the prescriptive analytics modules 152 for managing safety, before the steering actuator is actuated. Furthermore, see Kundu at [0095] which discloses that data analytics platform 145, the descriptive analytics module(s) 148 may decrypt the received vehicle data such as by using cryptographic hash algorithms such as MD5, SHA-1, SHA256, or other decryption techniques and that following decryption, the descriptive analytics module(s) 148 may authenticate or otherwise ***determine the identity of the vehicle and an occupant, and may access the vehicle data database 162 to retrieve any information*** about the vehicle or occupant maintained therein. Kundu at [0095] further discloses that the examples of the information that may be retrieved may include the vehicle sensor configuration information 128 and/or ***vehicle configuration information*** 130. Examiner further notes that determining the identity of the vehicle and an occupant corresponds to fulfilling the at least one safety condition. Also, see Kundu at [0119] which discloses that the service computing device 108 may determine vehicle dynamics for each candidate route and that FIG. 18 describes additional details of the vehicle dynamics algorithm, that for example, the vehicle dynamics algorithm may receive, as inputs, the candidate routes, road geometry from the map data database 156 database and vehicle chassis; see Kundu at [0120] which discloses that the service

computing device 108 may determine an optimal route based on the outputs of the route FOV coverage algorithm, drive horizon algorithm, and vehicle dynamics algorithm for the selected candidate routes selected by these three respective algorithms and that FIG. 19 describes additional details of the optimal routing algorithm, which may provide, ***as output, a route selected as the safest***, most energy efficient, and most comfortable ***route***. Alternatively, the Examiner may also map providing a route selected as the safest to checking whether the at least one safety condition is fulfilled. Alternatively, decrypting received information from the vehicle and identifying and authenticating the vehicle and/or occupant may correspond to checking whether the at least one safety condition is fulfilled. Examiner notes that the specification at [0043] discloses that the at least one safety condition is provided to be an element selected from the ***following group of safety conditions***: ***positive identity check of the motor vehicle and/or of the infrastructure***, presence of a predetermined safety integrity level (SIL or - automotive safety integrity level - ASIL) in the motor vehicle and/or the infrastructure, presence of a predetermined safety integrity level in one or in a plurality of communication links between the motor vehicle and the infrastructure, presence of a predetermined safety integrity level in a communication component for establishing the communication link between the motor vehicle and the infrastructure, presence of a predetermined safety integrity level in the overall system comprising the motor vehicle and the infrastructure and …. Based on the foregoing, the Examiner has shown a teaching based on a broadest

reasonable interpretation of the claim language in light of what is written in the specification.)

and outputting the control signals generated in the generating step, the infrastructure data is only used for at least partially automated driving if the at least one safety condition is fulfilled (see Kundu at [0021] which discloses that the vehicle may receive from the data analytics platform, information about one or more optimal routes selected by the data analytics platform for reaching the destination location, with the optimization performed by the data analytics platform being at least in part on selecting a route that is determined to be safe and fuel-efficient; Examiner notes that the one or more optimal routes are ***selected only when a route is determined to be safe***. Also, see Kundu at [0041] which discloses that the vehicle control program 124 may ***send control signals to the suspension controller, the steering controller, and/or the vehicle speed controller*** for controlling or partially controlling the vehicle in some applications; Examiner maps sending control signals to outputting the control signals. Furthermore, see Kundu at [0056] which discloses that upon determining the optimal route(s), the service computing device 108 may send the selected optimal route(s) 186 to the vehicle computing device 104; see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for ***partially/fully autonomous vehicles***, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, ***infrastructure cameras and other sensors***,

cellphones, other transportation data services, and so forth, as discussed above; see Kundu at [0058] which discloses that the data analytics platform 145 may ***provide road anomaly and driver behavior information for analyzing and making determinations with respect to vehicle occupant safety*** and that the data analytics platform may share aggregated vehicle data. Furthermore, see Kundu at [0110] which discloses that at 1304, the service computing device 108 may decrypt the received information and identify and authenticate the vehicle and/or occupant and as one example, the service computing device may ***first perform authentication to verify the connected vehicle and/or vehicle occupant.*** Examiner notes that performing the foregoing authentication may correspond to fulfilling the at least one safety condition.)

Kundu does not expressly disclose **is a presence of a predetermined safety integrity level (SIL or automotive safety integrity level ASIL) in the motor vehicle and in an infrastructure transmitting the infrastructure data,** which in a related art, Das teaches (see Das at [0002] which discloses that systems and control methods for ***autonomous or self-driving motor vehicles*** are known; see Das at [0003] which discloses that ASILs ("automotive safety integrity levels"), define ***safety and redundancy requirements*** for automobile technical systems depending on a hazard analysis and risk assessment in the case of a possible malfunction of the respective systems. Further, see Das at [0021] which discloses that one advantage in this case is that the operating safety of the control system can be increased by the ***reciprocal monitoring*** of the control units

by the ***monitoring unit***; see Das at [0022] which discloses that the control system can have a ***plurality of reciprocally checking control units***, which are also checked respectively by the control monitoring unit and that ***the ASIL of the control monitoring unit is the same as or higher than the ASIL of the respective control units*** in this case. Examiner notes that safety requirements may correspond to the at least one safety condition. Examiner notes that one or more of the monitoring unit, control units, and/or checking control units may correspond to the recited infrastructure transmitting the infrastructure data. Examiner notes that the presence of safety and redundancy requirements and reciprocal monitoring as well as the presence of ASIL in the control monitoring unit and the respective control units corresponds to the presence of SIL or ASIL in the motor vehicle and in the infrastructure. Examiner maps autonomous or self-driving motor vehicles to the recited motor vehicle. Examiner has shown a teaching based on a broadest reasonable interpretation of the claimed language.)

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Kundu to include is a presence of a predetermined safety integrity level (SIL or automative safety integrity level ASIL) in the motor vehicle and in an infrastructure transmitting the infrastructure data, as taught by Das.

One would have been motivated to make such a modification to avoid possible serious consequences by way of performing risk assessment and hazard analysis, in the

event of a possible malfunction of the respective systems, as suggested by Das at [0003-0004].

Regarding claim 2, the modified Kundu teaches the computer-implemented method of claim 1, **wherein a plurality of position signals representing a position of the motor vehicle are received and determining that a need exists for infrastructure-based, at least partially automated driving of the motor vehicle is performed based on the position of the motor vehicle** (see Kundu at [0059] in conjunction with Fig. 1 which discloses that the route selection program 126 may ***determine the vehicle's current location*** from the onboard sensors 112 such as via a GPS receiver or the like and that accordingly, the route selection program 126 may be executed ***to transmit information about vehicle's current location***, onboard sensor configuration information 128, and vehicle configuration information 130 such as, powertrain, trim level, etc. to the data analytics platform 145. Examiner maps vehicle's current location to position of the motor vehicle.)

Regarding claim 4, the modified Kundu teaches the computer-implemented method of claim 1, **wherein the infrastructure data are received from the infrastructure server via the communication link** (see Kundu at Fig. 1 which depicts ***network 106 communicatively coupled between vehicle computing device(s) and service computing device(s)***; Examiner maps one or more of service computing device(s)

to infrastructure server. Also, see Kundu at [0093] which discloses that the service computing device(s) 108 hosting the data analytics platform 145 may receive various types of information from various different sources and ***also may provide data to one or more of the sources and that the infrastructure information 1202 may include infrastructure camera images, and other information about infrastructure, road conditions, construction projects, and the like***.)

Regarding claim 5, the modified Kundu teaches the computer-implemented method of claim 1, **wherein the infrastructure data are checked and the control signals are generated based on a result of checking the infrastructure data** (see Kundu at [0041] which discloses that the vehicle control program 124 may use ***rule-based and or artificial-intelligence-based control algorithms to determine parameters for vehicle control***, and that for instance, the vehicle control program 124 may determine an appropriate action, such as braking, steering, accelerating, or the like, and may ***send one or more control signals*** to one or more vehicle systems 114 based on the determined action; see Kundu at [0048] which discloses modules that perform communications, encryption/decryption, data filtering, data fusion, and candidate route prediction and monitoring. Examiner notes that performing communications including encryption and decryption corresponds to checking data such as infrastructure data and applying a rule-based control algorithm to send one or more control signals corresponds to generating control signals based on the result of checking. Examiner has interpreted the claim

based on a broadest reasonable interpretation in light of what is written in the specification.)

Regarding claim 6, the modified Kundu teaches the computer-implemented method of claim 1, **wherein a plurality of motor vehicle data signals representing a plurality of motor vehicle data generated by the motor vehicle are received, and the infrastructure data is fused with the motor vehicle data to determine a plurality of infrastructure motor vehicle data, the control signals are generated based on the infrastructure motor vehicle data** (see Kundu at Fig. 12 at element 1224 (Authenticate, Filter, ***Fuse Data***); see Kundu at [0099] which discloses that the data analytics platform 145 may use data filtering and ***data fusion*** to ingest various types of timeseries and image data ***obtained from traffic infrastructure, user smartphones, third parties, and so forth*** and that the data may be ingested and stored in the databases 154 or the like. Examiner has shown a teaching based on a broadest reasonable interpretation of the claimed language in light of what is written in the specification.)

Claim 16 is directed toward a device that performs the steps recited in the computer-implemented method of claim 1. The cited portions of the reference used in the rejection of claim 1 teach the steps recited in the device of claim 16. Therefore, claim 16 is rejected under the same rationale used in the rejection of claim 1.

Claim 17 is directed toward a computer program that performs the steps recited in the computer-implemented method of claim 1. The cited portions of the reference

used in the rejection of claim 1 teach the steps recited in the computer program of claim 17. Therefore, claim 17 is rejected under the same rationale used in the rejection of claim 1.

Claim 18 is directed toward a machine-readable storage medium that performs the steps recited in the computer-implemented method of claim 1. The cited portions of the reference used in the rejection of claim 1 teach the steps recited in the machine-readable storage medium of claim 18. Therefore, claim 18 is rejected under the same rationale used in the rejection of claim 1. Also, see Kundu at [0031-0036] which discloses computer-readable media 118 and that the computer-readable media 118 may be used to store any number of functional components that are executable by the processor(s) 116.

Regarding independent claim 10, Kundu teaches a **computer-implemented method for infrastructure-based support of a motor vehicle driven in an at least partially automated manner, comprising the following steps: receiving a plurality of log-on data over a communication network to log on the motor vehicle with a remote infrastructure server and establish a communication link between the infrastructure server and the motor vehicle, the infrastructure data is sent from the infrastructure server over the communication link, it is checked during log in whether at least one safety condition for the infrastructure-based, at least partially automated driving of**

the motor vehicle is fulfilled, the at least one safety condition [is a presence of a predetermined safety integrity level (SIL or automotive safety integrity level ASIL) in the motor vehicle and in an infrastructure transmitting the infrastructure data**]** (see at least Kundu at [0038-0039] in conjunction with Fig. 1 which discloses that the ***one or more networks 106*** may include any appropriate network, including a wireless network, such as a cellular network; a wide area network, such as the Internet; a local area network, such an intranet; a local wireless network, such as Wi-Fi; close-rang wireless communications, such as BLUETOOTH® or DSRC; a wired network, including fiber optics and Ethernet; any combination of the foregoing, or ***any other suitable communication network***. See Kundu at Fig. 1 elements 102, 104; Examiner maps element 102 to the motor vehicle. See Kundu at Fig. 1 element 108; Examiner maps service computing device(s) 108 to remote infrastructure server. See Kundu at [0095] which discloses that data analytics platform 145, the descriptive analytics module(s) 148 may decrypt the received vehicle data such as by using cryptographic hash algorithms such as MD5, SHA-1, SHA256, or other decryption techniques and that following decryption, the descriptive analytics module(s) 148 ***may authenticate*** or otherwise determine the identity of the vehicle and an occupant, ***and may access the vehicle data database 162 to retrieve any information*** about the vehicle or occupant maintained therein. Kundu at [0095] further discloses that the examples of the information that may be retrieved may include the vehicle sensor configuration information 128 and/or ***vehicle***

configuration information 130. Examiner notes that Kundu teaches retrieving ***any information (i.e., while authenticating or logging in)***, such as the identity of the vehicle and the occupant, by way of accessing the vehicle data database ***during authentication or log in***. Examiner maps authentication to log in. Examiner further notes that determining the identity of the vehicle and an occupant corresponds to checking whether at least one safety condition for the infrastructure-based, at least partially automated driving of the motor vehicle is fulfilled.)

receiving a request to send a plurality of infrastructure data at the infrastructure server via the communication network, based on which the motor vehicle may be guided in an at least partially automated manner (see Kundu at Figure 1, which illustratively depicts network 106 for communicating a request between vehicle computing device(s) 104 and service computing device(s) 108; see Kundu at [0028] which discloses that implementations herein may significantly improve the performance of a fully/***partially autonomous vehicle*** using connected data and that the connected vehicle is able to share data with the data analytics platform, which provides access to cloud databases and cloud computing power, Web-based sources of information, as well as providing access to information provided by other vehicles; see Kundu at [0043] which discloses that the ***service computing device(s) 108 may include one or more servers*** or other types of computing devices that may be embodied in any number of ways, that for instance, in the case of a server, the programs, other functional

components, and data may be implemented on a single server, a cluster of servers, a server farm or data center, a cloud-hosted computing service, and so forth, although other computer architectures may additionally or alternatively be used; Examiner maps one of the one or more servers to the infrastructure server. Also, see Kundu at [0055] which discloses that the vehicle computing device 104 may ***request a route*** from the service computing device 108; Examiner notes that a request is made based on the need for automated driving or route guidance by the vehicle computing device(s). Also, see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for ***partially/fully autonomous vehicles***, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, ***infrastructure cameras and other sensors***, cellphones, other transportation data services, and so forth, as discussed above. Examiner maps the route requested by the vehicle computing device(s), which utilize various different types of data from different sources such as vehicles, infrastructure cameras, and other sensors, to the recited infrastructure data.)

generating the infrastructure data is generated by the infrastructure server based on a result of the check whether the at least one safety condition for the infrastructure-based supporting of the at least partially automated guided motor vehicle is fulfilled (see Kundu at [0021] which discloses that in some examples, the vehicle may access a connected ***data analytics platform*** such that the vehicle may

receive, from the data analytics platform, information about one or more optimal routes selected by the data analytics platform for reaching the destination location; see Kundu at Fig. 1, which depicts that data analytics platform 145 that resides in the service computing device(s) 108 (which may comprise one or more servers); see Kundu at [0048] which discloses that examples of predictive analytics modules 150 may include destination prediction, candidate route prediction and monitoring, speed profile determination, and anomaly prediction, that examples of ***prescriptive analytics modules 152 may include modules for managing safety,*** efficiency, comfort, and the like of vehicles and/or vehicle occupants; see Kundu at [0071] which discloses that in some examples herein, rather than relying on driver response when lane departure occurs, the LKA system may employ sensor fusion from the long-range camera and the long-range radar to alert the driver and also ***activate the steering actuator***, that accordingly, the steering actuator may be ***automatically engaged to return the vehicle to its proper lane***, and that ***sensor fusion algorithms*** may be required to meet strict performance and ***safety requirements***; for example, the Examiner maps activating a steering actuator to automatically engage to return the vehicle to its proper lane to fulfilling the at least one safety condition. Examiner notes that sensor fusion data associated with the long-range camera and the long-range radar is a provided to the data analytics platform which is checked by the prescriptive analytics modules 152 for managing safety, before the steering actuator is actuated. Also, see Kundu at [0110]

which discloses that at 1304, the service computing device 108 may decrypt the received information and identify and authenticate the vehicle and/or occupant and as one example, the service computing device may ***first perform authentication to verify the connected vehicle and/or vehicle occupant;*** Examiner further notes that performing authentication to verify the connected vehicle and/or vehicle occupant also corresponds to fulfilling the at least one safety condition. Also, see Kundu at [0119] which discloses that the service computing device 108 may determine vehicle dynamics for each candidate route and that FIG. 18 describes additional details of the vehicle dynamics algorithm, that for example, the vehicle dynamics algorithm may receive, as inputs, the candidate routes, road geometry from the map data database 156 database and vehicle chassis; see Kundu at [0120] which discloses that the service computing device 108 may determine an optimal route based on the outputs of the route FOV coverage algorithm, drive horizon algorithm, and vehicle dynamics algorithm for the selected candidate routes selected by these three respective algorithms and that FIG. 19 describes additional details of the optimal routing algorithm, which may provide, ***as output, a route selected as the safest***, most energy efficient, and most comfortable ***route***. Examiner may map providing a route selected as the safest to checking whether the at least one safety condition is fulfilled. Alternatively, decrypting received information from the vehicle and identifying and authenticating the vehicle and/or occupant may correspond to checking whether the at least one safety condition is fulfilled. Examiner

notes that the specification at [0043] discloses that the at least one safety condition is provided to be an element selected from the ***following group of safety conditions***: positive identity check of the motor vehicle and/or of the infrastructure, presence of a predetermined safety integrity level (SIL or - automotive safety integrity level - ASIL) in the motor vehicle and/or the infrastructure, presence of a predetermined safety integrity level in one or in a plurality of communication links between the motor vehicle and the infrastructure, presence of a predetermined safety integrity level in a communication component for establishing the communication link between the motor vehicle and the infrastructure, presence of a predetermined safety integrity level in the overall system comprising the motor vehicle and the infrastructure and …. Based on the foregoing, the Examiner has shown a teaching based on a broadest reasonable interpretation of the claim language in light of what is written in the specification.)

and transmitting the infrastructure data from the infrastructure server in response to receiving the request via the communication network, based on which the motor vehicle is driven in the at least partially automated manner, the infrastructure data is only transmitted to the motor vehicle if the at least one safety condition is fulfilled (see Kundu at [0021] which discloses that the vehicle may receive from the data analytics platform, information about one or more optimal routes selected by the data analytics platform for reaching the destination location, with the optimization performed by the data analytics platform being at least in part on selecting a route that

is determined to be safe and fuel-efficient; Examiner notes that the one or more optimal routes are ***selected only when a route is determined to be safe***. Furthermore, see Kundu at [0056] which discloses that upon determining the optimal route(s), the service computing device 108 may send the selected optimal route(s) 186 to the vehicle computing device 104; see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for ***partially/fully autonomous vehicles***, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, ***infrastructure cameras and other sensors***, cellphones, other transportation data services, and so forth, as discussed above; see Kundu at [0058] which discloses that the data analytics platform 145 may ***provide road anomaly and driver behavior information for analyzing and making determinations with respect to vehicle occupant safety*** and that the data analytics platform may share aggregated vehicle data. Furthermore, see Kundu at [0110] which discloses that at 1304, the service computing device 108 may decrypt the received information and identify and authenticate the vehicle and/or occupant and as one example, the service computing device may ***first perform authentication to verify the connected vehicle and/or vehicle occupant.*** Alternatively, the Examiner maps decrypting received information from the vehicle and identifying and authenticating the vehicle and/or occupant to checking whether the at least one safety condition is fulfilled, prior to transmission of the infrastructure data or selected optimal routes. Examiner maps the

sending of the selected optimal route(s) which incorporate the various sensor data to transmitting the infrastructure data in response to receiving the request.)

receiving the infrastructure data at the motor vehicle; (see Kundu at [0056] which discloses that upon determining the optimal route(s), the service computing device 108 may send the selected optimal route(s) 186 to the vehicle computing device 104; see Kundu at [0057] which discloses that to realize benefits of connected vehicle technologies for partially/fully autonomous vehicles, the connected data analytics platform 145 may receive various different types of the data from different sources such as vehicles 102, infrastructure cameras and other sensors, cellphones, other transportation data services, and so forth, as discussed above; see Kundu at [0058] which discloses that the data analytics platform 145 may provide road anomaly and driver behavior information for analyzing and making determinations with respect to vehicle occupant safety and that the data analytics platform may share aggregated vehicle data.)

generating a plurality of control signals for at least partially automated controlling of a lateral and/or a longitudinal operation of the motor vehicle based on the infrastructure data and (see Kundu at [0041] which discloses that the vehicle control program 124 may determine an appropriate action, such as braking, steering, accelerating, or the like, and may send one or more control signals to one or more vehicle systems 114 based on the determined action. Also, see Kundu at [0041], which

discloses that for example, the vehicle control program 124 may send control signals to the suspension controller, the steering controller, and/or the vehicle speed controller for controlling or partially controlling the vehicle in some applications. Examiner notes that control signals sent to the steering controller and/or the vehicle speed controller corresponds to controlling of a lateral and/or longitudinal operation of the motor vehicle.)

controlling the motor vehicle with the control signals generated in the generating step (see Kundu at [0041] which discloses that for example, the vehicle control program 124 may ***send control signals*** to the suspension controller, the steering controller, and/or the vehicle speed controller ***for controlling or partially controlling the vehicle*** in some applications.)

Kundu does not expressly disclose **is a presence of a predetermined safety integrity level (SIL or automative safety integrity level ASIL) in the motor vehicle and in the infrastructure**; which in a related art Das teaches (see Das at [0002] which discloses that systems and control methods for ***autonomous or self-driving motor vehicles*** are known; see Das at [0003] which discloses that ASILs ("automotive safety integrity levels"), define ***safety and redundancy requirements*** for automobile technical systems depending on a hazard analysis and risk assessment in the case of a possible malfunction of the respective systems. Further, see Das at [0021] which discloses that one advantage in this case is that the operating safety of the control system can be

increased by the ***reciprocal monitoring*** of the control units by the ***monitoring unit***; see Das at [0022] which discloses that the control system can have a ***plurality of reciprocally checking control units***, which are also checked respectively by the control monitoring unit and that ***the ASIL of the control monitoring unit is the same as or higher than the ASIL of the respective control units*** in this case. Examiner notes that safety requirements may correspond to the at least one safety condition. Examiner notes that one or more of the monitoring unit, control units, and/or checking control units may correspond to the recited infrastructure. Examiner notes that the presence of safety and redundancy requirements and reciprocal monitoring as well as the presence of ASIL in the control monitoring unit and the respective control units corresponds to the presence of SIL or ASIL in the motor vehicle and in the infrastructure. Examiner maps autonomous or self-driving motor vehicles to the recited motor vehicle. Examiner has shown a teaching based on a broadest reasonable interpretation of the claimed language.)

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Kundu to include is a presence of a predetermined safety integrity level (SIL or automative safety integrity level ASIL) in the motor vehicle and in the infrastructure, as taught by Das.

One would have been motivated to make such a modification to avoid possible serious consequences by way of performing risk assessment and hazard analysis, in the

event of a possible malfunction of the respective systems, as suggested by Das at [0003-0004].

Regarding claim 12, the modified Kundu teaches the computer-implemented method of claim 10, **wherein the infrastructure data is sent via the communication link only after the motor vehicle has successfully logged on to the infrastructure server** (see Kundu at [0110] in conjunction with Fig. 13 which illustratively depicts a process where at step 1304, the service computing device 108 may decrypt the received information and identify and authenticate the vehicle and/or occupant and that after authentication is successful, the other information obtained from the vehicle may be checked. Examiner maps the successful authentication of the vehicle and/or occupant to a successful logon. Furthermore, Kundu at [0121] in conjunction with Fig. 13 discloses that at step 1326, the service computing device 108 may send the selected optimal route to the vehicle and that for example, the vehicle may receive the selected optimal route, and may proceed along the selected route. Examiner notes that successful authentication is performed before the selected optimal route is sent.)

Regarding claim 19, the modified Kundu teaches the computer-implemented method of claim 1, **wherein determining that the need exists for infrastructure-based, at least partially automated driving of the motor vehicle is based on detecting whether the motor vehicle is in a predetermined traffic situation** (see Kundu at [0061]

which discloses that concept of autonomous driving mainly starts from "Level 3" ***(conditional driving automation***), in which the vehicle itself may ***monitor the surroundings and may exert some control over the vehicle*** (e.g., autonomous parking) and that there is no need for pedals or a steering wheel, as the autonomous vehicle system controls all critical tasks, monitors the surroundings, and ***identifies unique driving conditions***, such as ***traffic jams***, obstacles, road closures, and so forth. Also, see Kundu at [0093] in conjunction with Fig. 12 which discloses that the service computing device(s) 108 hosting the data analytics platform 145 may receive various types of information from various different sources and also may provide data to one or more of the sources, that examples include ***infrastructure information 1202***, user computing device instructions 1204, CAY sensor data 1206, ***travel demand information 1208***, map provider information 1210, OEM information 1212, and government entity information 1214, and that as mentioned above, the infrastructure information 1202 may include infrastructure camera images, ***and other information about infrastructure, road conditions***, construction projects, and the like. Further, see Kundu at [0115] which further discloses that the ***real-time traffic information*** may be updated such as periodically using time loop that executes at fixed time intervals or through any of various other update techniques, that the service computing device 108 may obtain the traffic data from a third party webserver, or the like and that the traffic information may be ingested in the database and ***sent to the routing and monitoring algorithm***

periodically. Examiner notes that the identification of unique driving conditions, such as traffic jams corresponds to detecting whether the motor vehicle is in a predetermined traffic situation.)

Regarding claim 20, the modified Kundu teaches the computer-implemented method of claim 19, **wherein the predetermined traffic situation is one of road work, a bridge, a freeway access road, a freeway interchange, a dangerous and/or a complicated road section, a traffic jam, a traffic circle, a bus stop, and a parking lot** (see Kundu at [0061] which discloses that concept of autonomous driving mainly starts from "Level 3" ***(conditional driving automation***), in which the vehicle itself may ***monitor the surroundings and may exert some control over the vehicle*** (e.g., autonomous parking) and that there is no need for pedals or a steering wheel, as the autonomous vehicle system controls all critical tasks, monitors the surroundings, and identifies unique driving conditions, such as ***traffic jams***, obstacles, road closures, and so forth.)

Regarding claim 21, the modified Kundu teaches the computer-implemented method of claim 1, **wherein the infrastructure data is a list of objects in an environment of the motor vehicle detected by a camera of the infrastructure** (see Kundu at [0049] which discloses: In addition, the computer-readable media 142 may store or access data used for performing the operations described herein. Further in some examples, the data may be stored in any suitable type data structures such as in one or more databases 154. Examples of databases 154 may include a ***map data***

database 156, a time series data database 158, an image data database 160, and a vehicle data database 162. For example, the map data database 156 may include information related to a required FOV for selected road segments, road profiles, high definition maps, and standard maps for various geographic regions…. Additionally, the image data database 160 may maintain ***images of roads, landmarks, intersections, and the like, such as may be received from infrastructure cameras***, cell phone cameras, vehicle-mounted cameras, and so forth. Examiner maps the images of ***roads, landmarks, intersections, and the like*** to list of objects in an environment of the motor vehicle. Examiner has shown a teaching based on a broadest reasonable interpretation of the claimed language.)

Claim 15 is rejected under 35 U.S.C. 103 as being unpatentable over Kundu et al. (US 2022/0065644) in view of Das et al. (US 2019/0283768) and further in view of Vassilovski et al. (US 2020/0236521).

Regarding claim 15, the modified Kundu teaches sending infrastructure data and using a broadcasting protocol (see Kundu at [0055-0058] [0094], for example), but does not expressly teach the computer-implemented method of claim 10, **wherein the infrastructure data is sent as a broadcast message or as a multicast message** which, in a related art, Vassilovski teaches (see Vassilovski at [0038] which discloses the use of V2X communication subsystem to generate a V2X notification and that the notification

may be a unicast message to a single receiver, a ***multicast message to multiple receivers, a broadcast message sent to all receivers within range, and/or any other suitable type of message***.)

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Kundu to include wherein the infrastructure data is sent as a broadcast message or as a multicast message, as taught by Vassilovski.

One would have been motivated to make such a modification to provide broadcast or multicast messages to multiple receivers or all receivers within a range, as suggested by Vassilovski at [0038].

Claim 22 is rejected under 35 U.S.C. 103 as being unpatentable over Kundu et al. (US 2022/0065644) in view of Das et al. (US 2019/0283768) and further in view of Spindler et al. (US 2012/0245848).

Regarding claim 22, the modified Kundu does not expressly disclose the computer-implemented method of claim 1, **wherein the infrastructure data is a plurality of objects positioned in a digital environment model representing an environment of the motor vehicle** which, in a related art, Spindler teaches (see Spindler at [0051] which discloses that: In one example, the navigation system may be configured ***to display the surrounding area of a determined location of the vehicle in a***

digital map with ***at least one item that represents a 3-dimensional object*** being displayed as a corresponding vector graphic in a 3-dimensional perspective view. Individual objects can be scanned by means of survey systems such as, for example, laser scanners, and digitized representations of the objects may be obtained. Based on the recorded pictures of buildings, or the like, 3-dimensional models may be synthesized as vector graphics. The displaying of individual objects in a 3-dimensional perspective view may improve the driver's orientation and help the driver to effectively and/or positively choose the correct road at a junction (i.e. the road to be taken to the predetermined destination.)

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Kundu to include wherein the infrastructure data is a plurality of objects positioned in a digital environment model representing an environment of the motor vehicle, as taught by Spindler.

One would have been motivated to make such a modification to display detailed digital maps indicating routes to destinations, the types of maneuvers to be taken at various locations such as, for example, junctions as well as different kinds of points of interest such as, for example, gas stations, restaurants, and landmarks, as suggested by Spindler at [0014].

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ROY RHEE whose telephone number is 313-446-6593. The examiner can normally be reached M-F 8:30 am to 5:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, Applicant may contact the Examiner via telephone or use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kito Robinson, can be reached on 571-270-3921. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ROY RHEE/
Primary Examiner, Art Unit 3664